ch16 Light and Sound

2012 Q19

19. Which of the following statements is **INCORRECT** ?

A. In air, the wavelength of infra-red radiation is shorter than that of ultra-violet radiation.
B. Visible light travels faster in air than in glass.
C. Microwaves travel at the speed of light in a vacuum.
D. Both light and sound exhibit diffraction.

2014 Q19

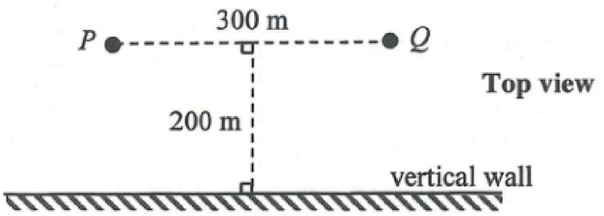
19. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are electromagnetic waves.
(2) Sound waves cannot travel in a vacuum.
(3) Sound waves cannot form stationary waves.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

2019 Q15

15.



Boys P and Q are 300 m apart and are both at a distance of 200 m from a vertical wall as shown. When P shouts once, two sounds are heard by Q. Which description below is correct ?
Given : speed of sound in air = 340 m s^{-1}

A. The first sound is louder and 0.59 s later the second sound is heard.
B. The first sound is louder and 0.29 s later the second sound is heard.
C. The second sound is louder and 0.59 s earlier the first sound is heard.
D. The second sound is louder and 0.29 s earlier the first sound is heard.

2020 Q21

21. Which of the following statements about ultrasound is/are correct ?

(1) Ultrasound has a shorter wavelength than audible sound.
(2) Ultrasound cannot be produced by vibrating objects.
(3) Ultrasound cannot be heard as it cannot travel through air.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2017 Q18

18. Two musical notes of the same pitch and loudness are produced by two different musical instruments. They sound different to the human ears because they have different

A. amplitudes.
B. phases.
C. wave speeds.
D. waveforms.

2019 Q22

22. Which of the following statements about infra-red radiation is/are correct ?

(1) It bends towards the normal when it travels from air to water.
(2) It travels faster in water than in air.
(3) It is used for satellite communication.

A. (1) only
B. (3) only
C. (1) and (2) only
D. (2) and (3) only

2017 Q21

21. If the speed of sound in water is x and the speed of light in water is y, which of the following is correct ?

	speed of sound in air	**speed of light in air**
A.	$> x$	$> y$
B.	$> x$	$< y$
C.	$< x$	$> y$
D.	$< x$	$< y$

2022 Q22

22. Which statements about ultrasound are correct ?

(1) Ultrasound is longitudinal waves.
(2) Ultrasound needs media to propagate.
(3) The speed of ultrasound in glass is higher than that in air.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

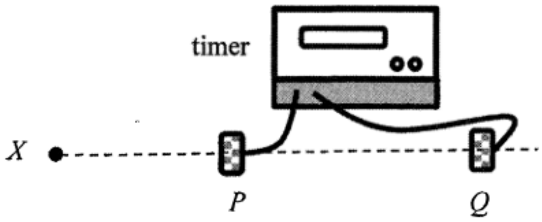
2012 Q23

23. Which of the following about ultrasound is **INCORRECT** ?

A. Ultrasound is a longitudinal wave.
B. The frequency of ultrasound is greater than 20000 Hz.
C. In air, the speed of ultrasound is faster than the speed of audible sound.
D. In air, the diffraction effect of ultrasound is less prominent than that of audible sound.

2017 Q15

15. An experiment is set up to measure the speed of sound in air as shown. P and Q are two microphones connected to a timer. A sound is produced at X. The timer starts when P receives the sound, and stops when Q receives the sound. The timer shows the time taken for the sound to travel from P to Q. The distance PQ and the time shown can be used to calculate the speed of sound.

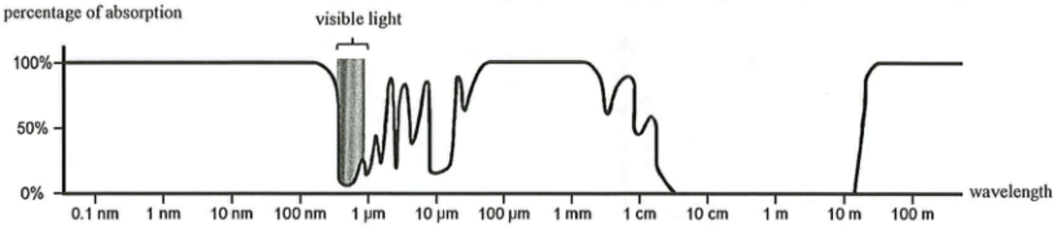


Which of the following statements is **INCORRECT** ?

A. X, P and Q must be along the same straight line.
B. The percentage error in the time measured will increase if the distance PQ is reduced.
C. The speed of sound determined should be independent of the distance between X and P.
D. The distance PQ must be equal to an integral multiple of wavelengths of the sound produced at X.

2023 Q22

22. When measuring electromagnetic waves from outer space on the Earth's surface, the percentage of absorption by the atmosphere for various wavelengths of electromagnetic waves is shown below.



Referring to this absorption graph, determine which statement below is correct.

A. Gamma rays from galaxies can be observed on Earth's surface.
B. Nearly all ultra-violet radiation can reach Earth's surface.
C. Infra-red radiation from stars can reach Earth's surface without absorption.
D. GHz radio waves (wavelength about 0.5 m) are the best for communications between ground station and space station in the upper atmosphere.

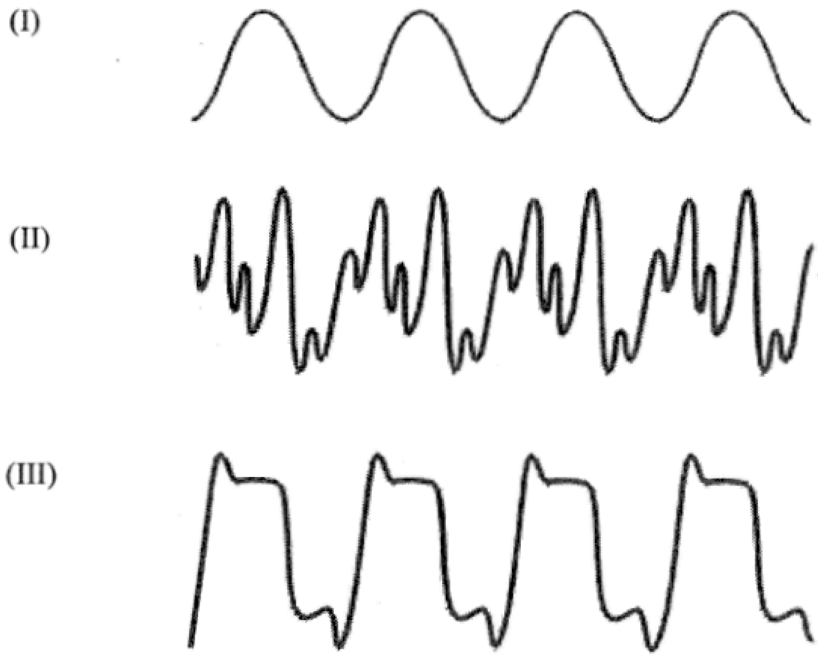
2021 Q19

19. Typical wavelengths of X-rays and microwaves are in the ratio $10^n : 1$. The value of n could be

A. −10.
B. −4.
C. +4.
D. +10.

2012 Q22

22. The figure shows the waveforms of sound notes generated by a violin, a piano and a tuning fork. The scale is the same in time and intensity axes for all three waveforms.



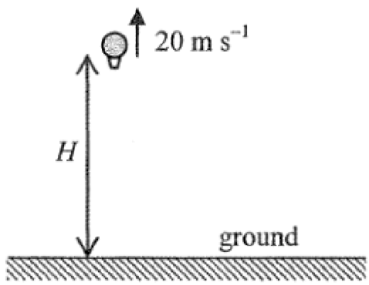
Which of the following about the sound notes are correct ?

(1) They all have the same pitch.
(2) The qualities of sound of (II) and (III) are different.
(3) (I) is generated by the tuning fork.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

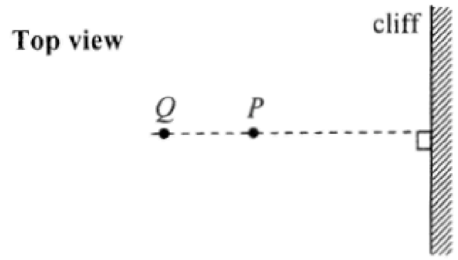
2015 Q19

19. A balloon is rising at a uniform speed of 20 m s^{-1}. When the balloon is at an altitude H as shown, it sends a sound signal towards the ground. After 5 s, the balloon receives the echo of the signal. Estimate H. Given: speed of sound in air = 340 m s^{-1}



A. 1600 m
B. 850 m
C. 800 m
D. 750 m

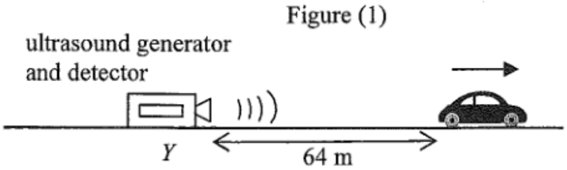
19.



Astronauts P and Q stand at 400 m and 600 m respectively from a vertical cliff on the surface of a planet. The figure shows the top view. P claps his hands once and Q hears two clapping sounds separated by 4 s. What is the speed of sound in the atmosphere of this planet ?

A. 100 m s^{-1}
B. 150 m s^{-1}
C. 200 m s^{-1}
D. 250 m s^{-1}

17.



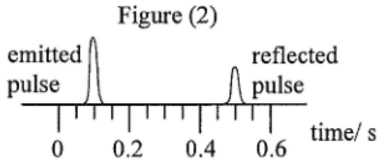


Figure (1) shows a car travelling with a uniform speed along a straight road away from a stationary ultrasound generator and detector at Y. When the car is 64 m from Y, the generator emits an ultrasound pulse towards the car. The pulse is then reflected back to the detector at Y and displayed on a CRO as shown in Figure (2). Estimate the speed of the car. Given: speed of ultrasound in air is 340 m s^{-1}

A. 16 m s^{-1}
B. 20 m s^{-1}
C. 24 m s^{-1}
D. 32 m s^{-1}

19. Which of the following phenomena provides conclusive evidence that sound is a wave ?

(1) reflection of sound from a wall
(2) refraction of sound at the boundary between two media
(3) interference of sound

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only

2015 Q20

20. Which of the following gives the order of magnitude of the wavelengths of ultra-violet radiation and microwave in a vacuum ?

	ultra-violet radiation	microwave
A.	10^{-8} m	10^{-2} m
B.	10^{-8} m	10^{-5} m
C.	10^{-10} m	10^{-2} m
D.	10^{-10} m	10^{-5} m

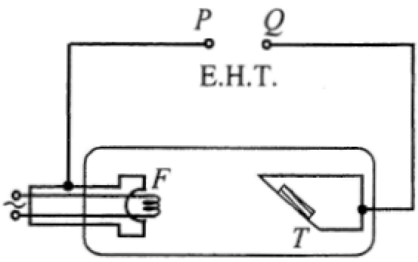
2018 Q21

21. Which of the following is **NOT** a typical sound intensity level that occurs in daily life ?

A. 130 dB : when an airplane take-off
B. 110 dB : at a rock concert
C. 80 dB : having a normal conversation
D. 30 dB : inside a library

2012 Q34

34.



The figure shows a schematic diagram of an X-ray tube in which the filament F and the metal target T are connected to terminals P and Q of an E.H.T. Which statement is correct ?

A. P is the positive terminal and X-rays are emitted from T.
B. P is the positive terminal and X-rays are emitted from F.
C. Q is the positive terminal and X-rays are emitted from T.
D. Q is the positive terminal and X-rays are emitted from F.

2021 Q20

20. Submarines employ ultrasound instead of microwaves to detect obstacles in the sea. This is because

A. wavelengths of ultrasound are shorter than those of microwaves.
B. ultrasound travels faster than microwaves in the sea.
C. microwaves are easily absorbed by sea water.
D. microwaves diffract too much in the sea.

2016 Q23

23. Which of the following are applications of ultrasound ?

(1) sterilizing drinking water
(2) detecting cracks in railway tracks
(3) breaking up kidney stones

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

2024 Q21

21. Which description below is **INCORRECT** ?

A. Microwaves are a kind of electromagnetic waves.
B. Microwaves can be observed with the naked eye.
C. Microwaves propagate with the speed of light in a vacuum.
D. Microwaves are used in radar to detect the position of an aeroplane.

2024 Q22

22. Figure (a) and (b) are the CRO displays of musical note *X* and *Y* using the same time axis and intensity axis.

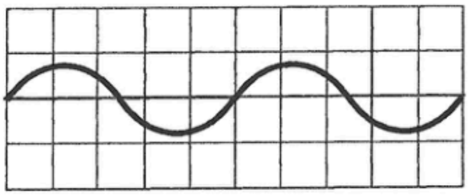
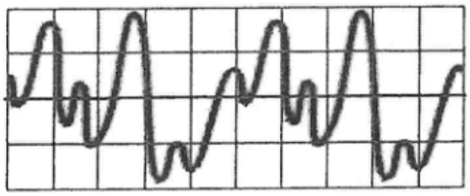
Figure (a)

note *X*

Figure (b)

note *Y*

Which comparison of these two notes must be correct ?

	pitch	loudness
A.	the same	the same
B.	the same	not the same
C.	higher for *Y*	the same
D.	higher for *Y*	not the same

pp Q16

16. Which of the following phenomena demonstrates that light is an electromagnetic wave ?

A. Light carries energy.
B. Light reflects when it meets a polished metal surface.
C. Light bends when it travels across a boundary from one medium into another.
D. Light can travel from the Sun to the Earth.

sap Q22

22. Which of the following statements about sound waves is/are correct ?

(1) Sound waves are longitudinal waves.
(2) Sound waves are electromagnetic waves.
(3) Sound waves cannot travel in a vacuum.

A. (2) only
B. (3) only
C. (1) and (2) only
D. (1) and (3) only